Exp1 toc docs

1. Write a C program to simulate a Deterministic Finite Automata (DFA) for the

given language representing strings that start with a and end with a

1. Aim

To write a C program to simulate a Deterministic Finite Automaton (DFA) that accepts strings over the alphabet ({a,b}) which start with a and end with a.

2. Procedure

1. Start the program.
2. Read the input string containing a and b.
3. Initialize the DFA to its initial state, q0.
4. Read each character of the input string one by one.
5. Change the state according to the DFA transition table.
6. After reading all characters, check the final state.
7. If the final state is q1, print String Accepted; otherwise, print String Rejected.
8. Stop the program.

Program:

```
#include <stdio.h>
#include <string.h>

int main() {
        char str[100];
        int i, state = 0;

        printf("Enter a string over {a, b}: ");
   scanf("%s", str);

        for (i = 0; str[i] != '\0'; i++) {
        if (state == 0) {
        if (str[i] == 'a')
                state = 1;
        else
```

```
            state = 3;
        }
        else if (state == 1) {
        if (str[i] == 'a')
            state = 1;
        else
            state = 2;
        }
        else if (state == 2) {
        if (str[i] == 'a')
            state = 1;
        else
            state = 2;
        }
        else {
        state = 3;
        }
        }

        if (state == 1)
    printf("String Accepted\n");
        else
    printf("String Rejected\n");

        return 0;
}
```

Result: Thus, the C program to simulate a DFA that accepts strings starting with a and ending with a was written and executed successfully.